User Manual

ANEH5-US4MBP Ultrasonic Anemometer

180322

BLET Measurement Group

1.FOREWORD

Thank you for purchasing the ANEH5-US4MBP manufactured by BLET.
To achieve optimum performance we recommend that you read the whole of this manual before proceeding with use.
BLET products are in continuous development and therefore specifications may be subject to change and design improvements without prior notice.
The information contained in this manual remains the property of BLET and must not be copied or reproduced for commercial gain.

Regulatory Compliance

ANEH5-US4MBP has CE certification of EU
The electromagnetic compatibility of the ANEH5-US4MBP has been tested in accordance with the following applicable directives:
LVD 2014/35/EU Low Voltage
EMC 2014/30/EU Electromagnetic Compatibility
EMC 2014/35/EU Electromagnetic Compatibility
Our above product has been assessed against the following applicable standards:
EN 61010-1:2010
EN 61326-1:2013
EN 61000-3-2:2014
EN 61000-3-3:2013
EN 60945-1:2003

Warranty

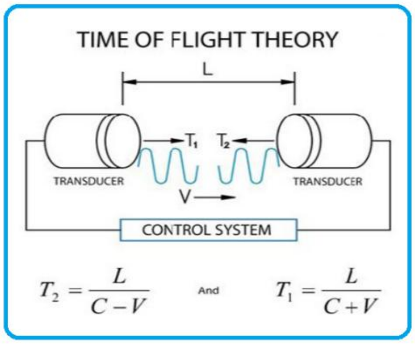
BLET hereby represents and warrants all Products manufactured by BLET and sold here under to be free from defects in workmanship or material during a period of twenty-four (24) months from the date of delivery save for products for which a special warranty is given.

2.INTRODUCTION

ANEH5-US4MBP ultrasonic anemometer is very robust with no moving parts, maintenance free , simultaneously output wind speed and direction. Each unit is factory calibrated in our wind-tunnel testing lab prior to shipping.
ANEH5-US4MBP powers up with 3 ... 30 VDC and outputs serial data with a selectable communication protocol: SDI-12, MODBUS-RTU and NMEA 0183. Four alternative serial interfaces are selectable: RS-232, RS-485 and SDI-12. The transmitter is equipped with a 4-pin M12 connector for installation.
The following options are available:
- Windows XP/WIN7 based testing software, which only support MODBUS protocol
- USB to RS-232/RS-485 converter cable (1.5m)
- Mounting kits

3.WORKING PRINCIPLE

Measure the transmission time of ultrasonic sensors from sensor N to sensor S, and compare with the transmission time of sensor S to sensor N. Similarly, compare the time of W to E and E to W time. (N = north, S = south, E = east, W = west)



For example, if the wind blew from the north, time of ultrasonic from N to S will be shorter than from S to N, and transmission time of it from W to E and E to W is the same. Through calculating the time difference of ultrasonic transmission between two points, the wind speed and direction can be calculated. This calculation method has nothing to do with other factors such as temperature.

The wind speed is represented as a scalar speed in units m/s. The wind direction is expressed in degrees (°). The wind direction reported indicates the direction that the wind comes from. North is represented as 0°, east as 90°, south as 180°, and west as 270°.

About measurement sampling rate: The ultrasonic probes sample hundreds of times per second, and process those raw data as wind speed and wind direction output every second.

4.TECHNICAL SPECIFICATION

	Range	Accuracy	Resolution
Wind Speed	0 - 40m/s	±5%	0.1m/s
Wind Direction	0 - 359°	±3 °	1°
Digital Output	RS485、RS232、SDI-12		
Analog Output	4-20mA(optional)		
Baud Rate	4800 - 19200 bps		
Communication Protocol	ModBus-RTU、NMEA-0183、SDI-12		
Protection Grade	IP65		
Operating Temperature	-40℃ - +60℃		
Storage Temperature	-50℃ - +80℃		
Working Humidity	0 - 100%		
Power Supply	VDC: 12-24V		
Power consumption	22mA @12V		
Dimension/Weight	ABS: Φ82×108mm 、0.28kg Aluminum alloy： Φ82×125mm 、0.38kg		
Material	ABS engineering plastic or aluminium alloy (cost extra $50)		

Specifications may be subject to change without prior notice.

6.PACKING LIST

Item	Quantity
ANEH5-US4MBP Anemometer	1 pcs
4m communication cable with watertight plug	1 pcs
User Manual	1 pcs

PACKING NOTICE

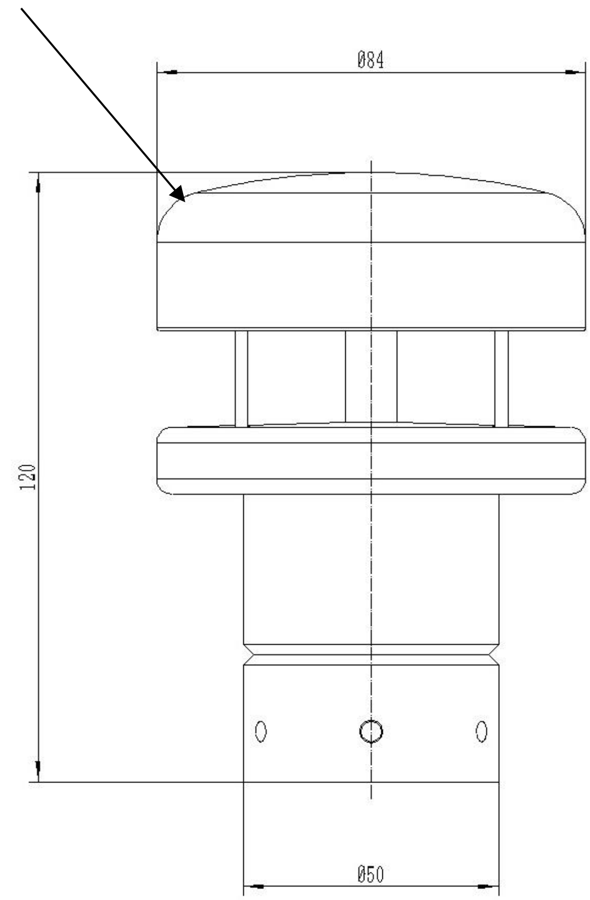
By keeping ANEH5-US4MBP well positioned in ex-work package when it’s beingtransferred to protect it from any potential damage.

7.APPERANCE SKETCH

7.1 Metal Shell

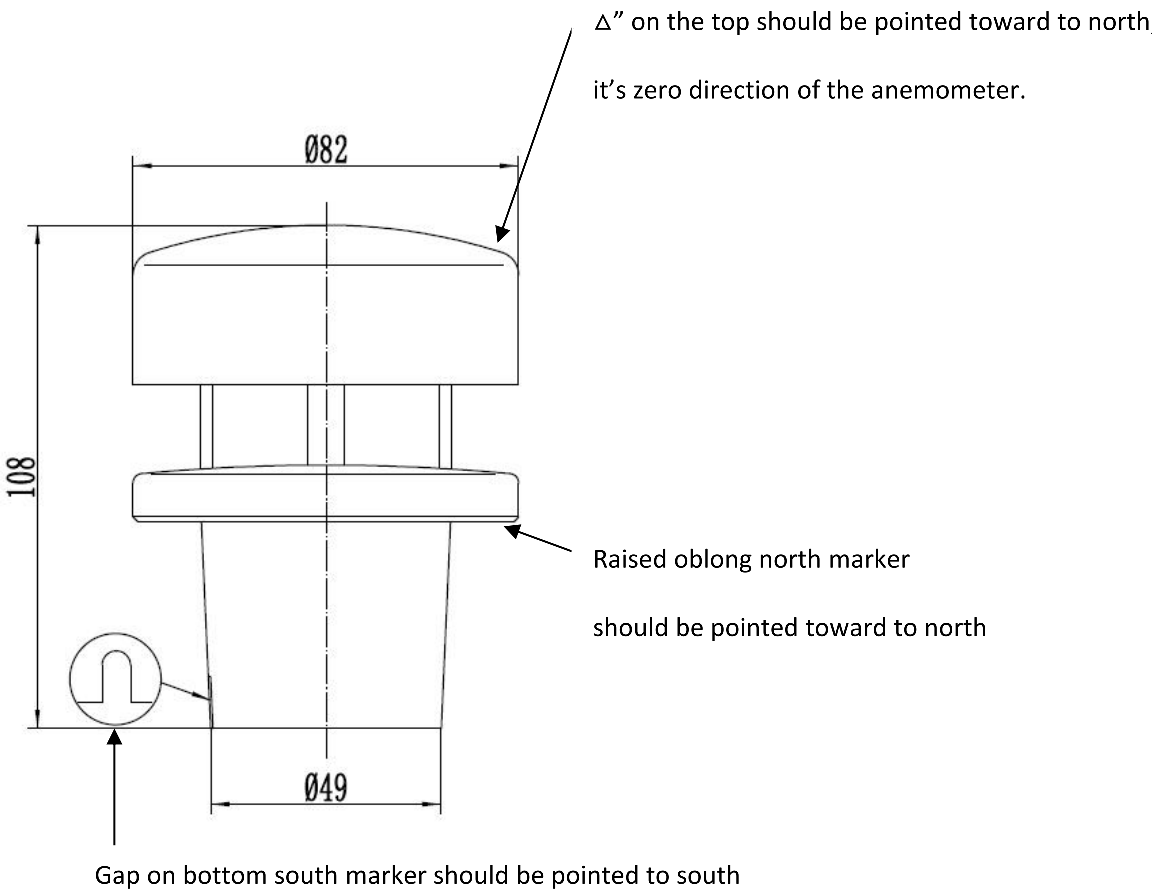
North marker”△” on the top help you align unit toward to north

It’s also zero direction of the anemometer.



7.2 ABS Shell

There has three north markers on ABS shell for easy observation from downward, front upward.



8.WIRING

For RS485 & 4-20mA

Yellow	Green	Grey	Black	White	Brown	Red	Blue
DA RS485	DB RS485	Wind Direction 4-20mA+	Wind Direction 4-20mA GND	Wind Speed 4-20mA+	Wind Speed 4-20mA GND	Power+	Power-
RS485		Wind Direction		Wind Speed		12-24VDC	GND of Power

Note:

1.Default output is RS485& 4-20mA.

OKFinal definition of cable wiring should be referred to sticker on cable.

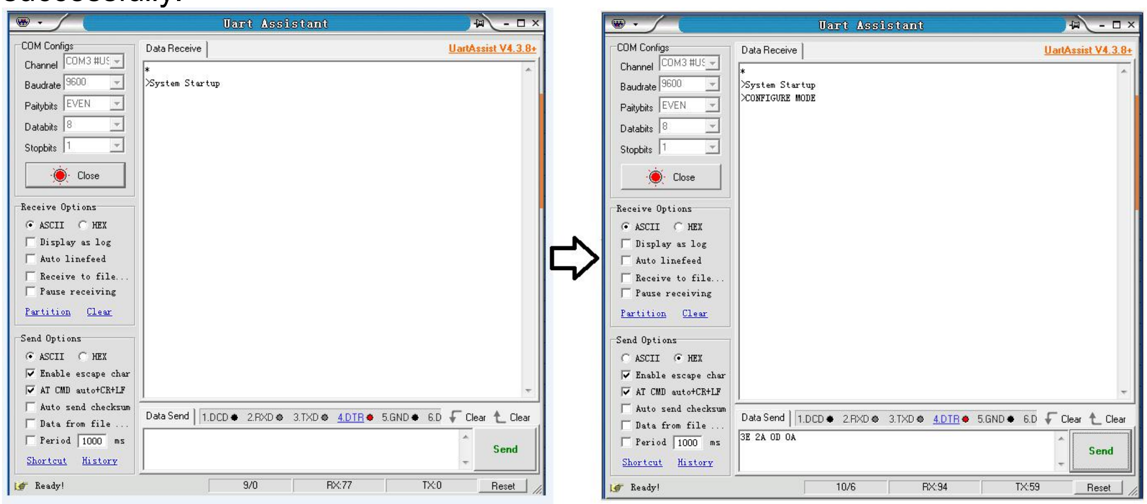
9.Procedure of confirmation of wiring and communication

3 seconds after wiring of our Device and correctly configuring serial communication tool, our instrument will output characters “>System Startup” in ASCII(0A 3E 53 79 73 74 65 6D 20 53 74 61 72 74 75 70 0D 0A in HEX), which indicate that our instrument is powered up.

We can simply test its response by inputting “enter setting mode” command “3E 2A 0D 0A”.

Our instrument will immediately respond “>CONFIGURE MODE” in ASCII(3E 43 4F 4E 46 49 47 55 52 45 20 4D 4F 44 45 0D 0A in HEX).

So far, the communication test is finished, device is proven to be communicated successfully.

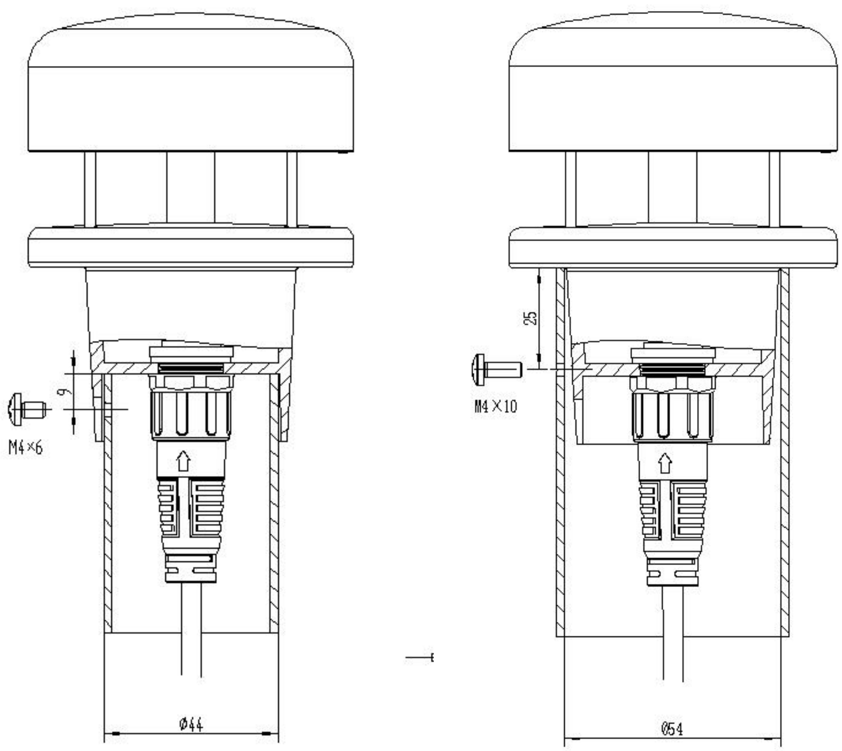


10.INSTALLATION GUIDELINES

10.1 The ANEH5-US4MBP can be mounted on top mast posts, poles, tripods, etc. Depending on the application. Pole should have three 3 spaced holes for M5 screw 7.5mm lower than top of pole equally plug cable(waterproof aerial plug) through pole. ANEH5-US4MBP should be mounted two to three meters above ground.

Note: The user must have proper stress relief on the cable. Turn the plug and press it gently into the socket to connect the plug to the ANEH5-US4MBP outlet. When the plug is connected, turn the outer sleeve clockwise and lock the plug. With 3 stainless steel screws, the ANEH5-US4MBP can be fixed to the mounting pipe (the screw has a maximum installed torque of 4Nm).

（refer to picture on the bottom）



Orientation: Use a standard compass to find correct geographic north direction then align north marker to it, then fix anemometer

10.2 You will need to adjust the anemometer so that it is level. Use a bubble level or other leveling device to ensure the anemometer is level (leveling device not included).

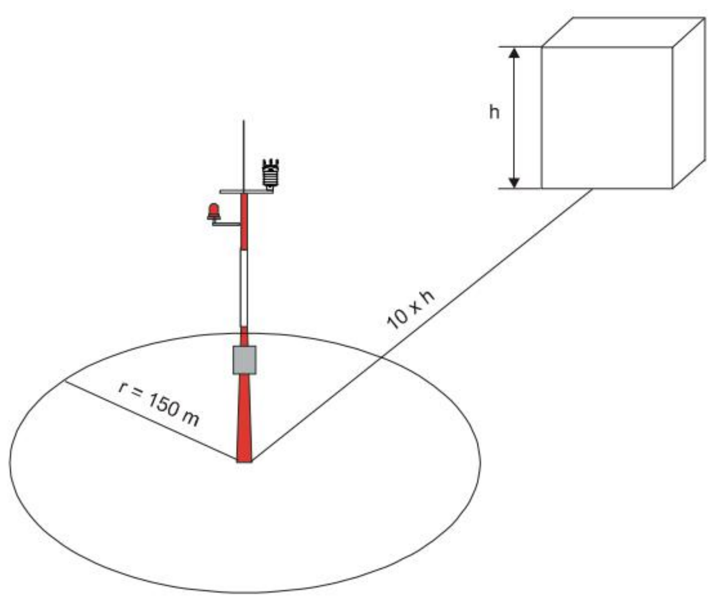
10.3 Customers must ensure that the ANEH5-US4MBP is installed in an open area so as to avoid obstacles to airflow or turbulence in the surrounding buildings.

Do not install ANEH5-US4MBP on the side of a high power radar or radio transmitter.

10.4 Installation Sketch

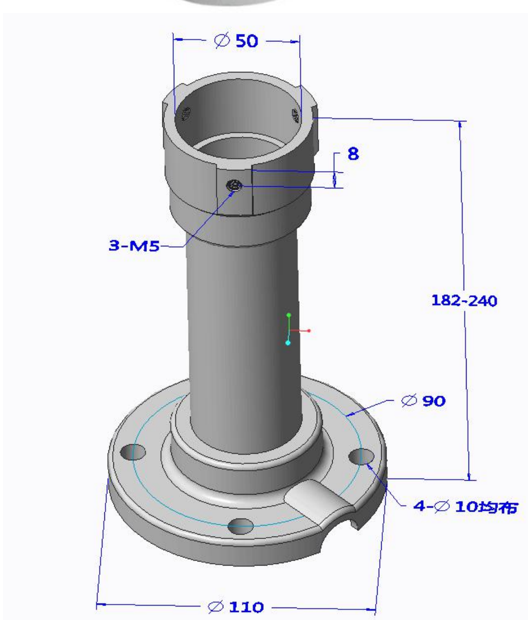
Following World Meteorological Organization (WMO) guidelines, a general recommendation is that there is at least 150 m open area in all directions from the mast. Any object of height (h) does not significantly disturb wind measurement at a minimum distance of 10 times the height of the object.

The recommended minimum length (h) for the mast that is installed on top of a building is 1.5 times the height of the building (H). When the diagonal (W) is less than the height (h),the minimum length of the mast is 1.5 W. However, follow the application specific instructions and local regulations when placing ANEH5-US4MBP.



11.Optional installation pole(Cost extra 20 USD/PCS)

In order to facilitate the installation for users, we specially customized a cast aluminum bracket for users' option, installation way shown as pictures below:



12.CLEANING

If there is any build up of deposit on the unit, it should be gently cleaned with a cloth moistened with soft detergent. Solvents should not be used, and care should be taken to avoid scratching any surfaces. The unit must be allowed to defrost naturally after being exposed to snow or icy conditions, do NOT attempt to remove ice or snow with a tool.

13.AFTER-SALE SERVICE

There are no moving parts or user-serviceable parts requiring routine maintenance. Opening the unit or breaking the security seal will void the warranty and the calibration. In the event of failure, prior to returning the unit to your authorised BLET distributor, it is recommended that:

1. All cables and connectors are checked for continuity, poor contacts, corrosion etc.
2. You contact your supplier for advice

If you ever need assistance with your ANEH5-US4MBP or have any questions or feedback, there are several ways to contact us. BLET has customer service representatives available to speak with you Monday through Friday, between 9 am and 6 pm UTC+8 time.

Note: If you purchased your anemometer through a distributor, please contact them for assistance.

Our email: technique@blet-mesure.fr

14.Calibration

The anemometer calibration is based on fundamental physical principles and does not change with use. Recalibration should therefore not be necessary.

15.INSTRUMENT RETURN

If the instrument needs to be returned, please carefully pack the instrument in the original package and deliver it to the authorized agent of the BLET with the detailed explanation of malfunction.

16.COMMUNICATION PROTOCOL

Refer to appendix.

Compact Weather station communication protocol

ModBus-RTU V1.08

Modbus Specification

Start Bit	1 bit
Data Bits	8 bit
Parity	EVEN
Stop Bits	1 bit
Baud Rate	9600 Baud

Communication interface

Communication interface:RS485 or RS232, default interface:***RS485***。

Communication Protocol

MODBUS Protocol - RTU Mode.

Protocol Description

MODBUS protocol defines a simple protocol data unit(PDU) independent from basic communication layer.

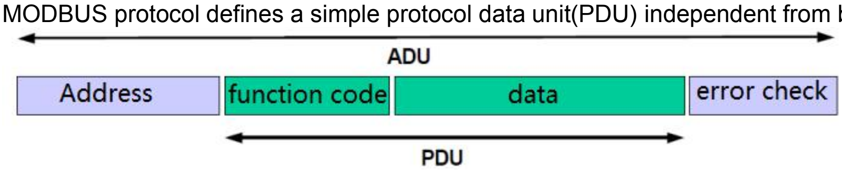


Figure 2. Commonly used MODBUS frame

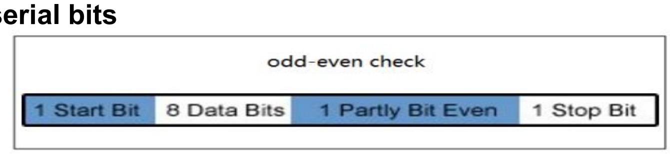
MODBUS has two transmission mode:RTU and ASCII.

Our sensor adopts RTU mode.

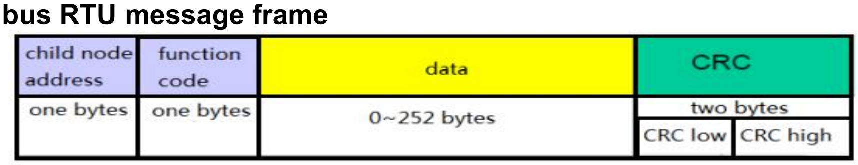
1. RTU transmission mode

When controllers are setup to communicate on a Modbus network using RTU (Remote Terminal Unit) mode, each eight-bit byte in a message contains two four-bit hexadecimal characters. The main advantage of this mode is that its greater character density allows better data throughput than ASCII for the same baud rate. Each message must be transmitted in a continuous stream.

- **RTU Mode serial bits**



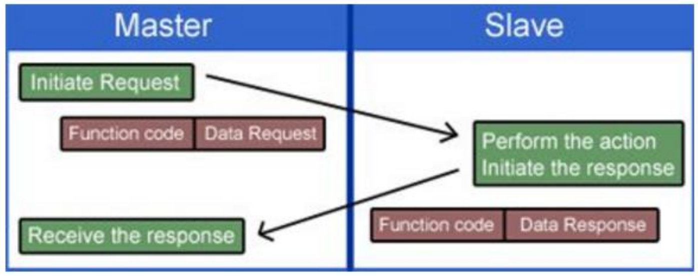
- **Modbus RTU message frame**



- **CRC Check**

  RTU Mode has Cyclical Redundancy Checking(CRC) on all content of message, no matter if there is an odd-even check or not.
  CRC check code is a 16 bits value composed by two 8 bits value and added as tail of message. After calculation, lower byte first then high byte. CRC higher byte is the last byte of message.
  The CRC check code is calculated by sender. Receiver will recalculate CRC check code and compare it with CRC code received, if they are different, then there is an error happen during transmission.

MODBUS communication Mode



Data Coding
MODBUS use "big-Endian" to indicate address and data, which means when there is several bytes be sent out, the most significant bit is sent and received first.

Eg.

Register Size	Value
16bit	0x1234

The first byte is 0x12, then 0x34

1. Protocol of Device

- **Function Code Supported**

Function code type	Length	Function Code（HEX）	Description
Data access	16 bit	03	Read data from internal register
Data access	16 bit	10	Write data to multiple register

- **Error Code Supported**

Error code	Description
01	Function Code Error
02	Register Address Error
03	Register Value error
06	Device Busy

- **Internal Registers Description:**

Register	Length	Data Type	Definition	Range
Register 1	16 bit	16 bit int	Device State	0x0000 ~0xFFFF Refer to Appendix I
Register 2	16 bit	16 bit int	Wind Direction	0 - 359 °
Register 3	16 bit	32 bit float	Wind Speed	0 - +60 m/s
Register 4	16 bit			
Register 5	16 bit	32 bit float	Temperature	-40 - +80 ℃
Register 6	16 bit			
Register 7	16 bit	32 bit float	Humidity	0 - 100 %
Register 8	16 bit			
Register 9	16 bit	32 bit float	Air Pressure	150 - 1100 hPa
Register 10	16 bit			
Register 11	16 bit	16 bit int	Compass Heading	0 - 359 °
Register 12	16 bit	16 bit int	Precipitation Type	Refer to Appendix II
Register 13	16 bit	32 bit float	Precipitation Intensity	Single-precision
Register 14	16 bit			
Register 15	16 bit	32 bit float	Accumulated Precipitation	Single-precision
Register 16	16 bit			
Register 17	16 bit	16 bit int	UV Index	UV index from 0 to 15
Register 18	16 bit	16 bit int	GPS Status	0: Positioned 1: No Positioned
Register 19	16 bit	32 bit float	GPS Speed	Km/h
Register 20	16 bit			
Register 21	16 bit	16 bit int	GPS Heading	0 - 359 °
Register 22	16 bit	32 bit float	Longitude	East: positive West: negative
Register 23	16 bit			
Register 24	16 bit	32 bit float	Latitude	North: positive South: negative
Register 25	16 bit			
Register 26	16 bit	32 bit float	PM2.5 concentration	0-500 ug/m^3
Register 27	16 bit			
Register 28	16 bit	32 bit float	Visibility	m
Register 29	16 bit			
Register 30	16 bit	32 bit float	Radiation Illuminance	Lux
Register 31	16 bit			
Register 32	16 bit	32 bit float	Accumulated solar radiation	Daily Solar Radiation KJ
Register 33	16 bit			
Register 34	16 bit	32 bit float	Solar Radiation Power	W/m^2
Register 35	16 bit			
Register 36	16 bit	32 bit float	Compass Corrected (True) Wind Direction	0 ~ 359.9 °
Register 37	16 bit			
Register 38	16 bit	32 bit float	Altitude	m
Register 39	16 bit			
Register 40	16 bit	32 bit float	GPS Corrected(True) Wind Speed	0-60m/s
Register 41	16 bit			
Register 42	16 bit	32 bit float	Accumulated Snow Thickness	m
Register 43	16 bit			
Register 44	16 bit	32 bit float	UV Radiation	W/m2
Register 45	16 bit			
Register 46	16 bit	32 bit float	PM1.0 concentration	0-500 ug/m^3
Register 47	16 bit			
Register 48	16 bit	32 bit float	PM10 concentration	0-500 ug/m^3
Register 49	16 bit			
Register 50	16 bit	32 bit float	Color Temp	K
Register 51	16 bit			

Note: Starting address of registers start from zero, E.g. address of register 1 is *0x0000*

- **32 bit float type format**

D3	D2	D1	D0
Higher byte	Middle byte 1	Middle byte 2	Lower byte

- **Format of data stored in register**

Definition	Register	Bit	Byte position
Wind speed	Register 2-higher byte	8 bit	D1
	Register 2-lower byte	8 bit	D0
	Register 3-higher byte	8 bit	D3
	Register 3-lower byte	8 bit	D2

- **Function code(0x03) description - read holding register**

A remote device can use function code to read data of holding register. The request PDU specifies starting address and quantity of registers. Register address from zero,therefore, the address register 1-3 corresponds to address 0-2.

Response packet from each register is divided into two bytes in binary format. The first byte is higher bits, the second byte is lower bits.

Request

Function Code	1 byte	0x03
Starting Address	2 bytes	0x0000 ~ 0x005F
Register Quantity	2 bytes	1 ~ 96

Response

Function Code	1 byte	0x03
Bytes Quantity	1 bytes	N*2
Register Data	N*2 bytes	

Note： N is quantity of registers

Error response

Error Code	1 byte	0x83
Exception Code	1 byte	01 or 02 or 03 or 06

Communication Example：

Read 96 internal registers

Communication Example：

Request		**Explanation**
Description	(HEX)	
Device address	01	Request instrument of "01" address for data of 52 registers starting from register Number "00". For instance: Address of our register start coding from 00~50 (Total quantity:51)
Function code	03	
Starting address higher byte	00	
Starting address lower byte	00	
Read register number higher byte	00	
Read register number lower byte	33	
Checksum higher byte	05	
Checksum lower byte	DF	

- **Request：**
  (HEX) 01030000003305DF
- **Response：**
  (HEX)

**0103C05DFF000063AC3CAE876441CC71B24289CFDD446C
002E000133333F3300000000000F000174BC3D130000EC764
2CFE59741F56666418A000000008A0046A9CCCD413C999A
436100004238F854440D63AC3CAE0000000047AE3FE1CCC
D412C000041980000000008279
(Detailed analysis is at next page)**

Explanation of above response string in HEX as below:

Instrument address		01	Address is "01"
Function Code		03	"03" means read
Total Bytes		C0	192 bytes
Register	1	5DFF	Device state:1011101111111 (from left to right) 1:UV radiation 0:Snow thickness 1:True wind speed 1:True wind direction 1:Altitude 0:Visibility 1:Luminance 1:Solar radiation 1:PM1.0/2.5/10 1:GPS 1:Precipitation 1:Compass 1:Pressure 1:Relative wind 1:Temperature and humidity
Register	2	0000	Wind direction: 0°
Register	3-4	63AC3CAE	Wind speed: 0.02 m/s Converted by 3CAE63AC all 32 bit float date in this protocol comply to IEEE754 Standard
Register	5-6	876441CC	Temperature: 25.57℃
Register	7-8	71B24289	Humidity: 68.7%
Register	9-10	CFDD446C	Pressure: 947.2 hPa
Register	11	002E	Compass heading: 46°
Register	12	0001	Precipitation type:001 means rain
Register	13-14	33333F33	Rain intensity: 0.7 mm/h
Register	15-16	00000000	Accumulated rain:0 mm
Register	17	000F	UV index: 15
Register	18	0001	GPS state: "1" means positioned
Register	19-20	74BC3D13	Traveling speed: 0.036 Km/h
Register	21	0000	Traveling heading: 0°
Register	22-23	EC7642CF	Longitude: 103.961838
Register	24-25	E59741F5	Latitude: 30.737104
Register	26-27	6666418A	PM2.5: 17 µg/m3
Register	28-29	00000000	Visibility(reserved)
Register	30-31	8A0046A9	Luminance: 21701 Lux
Register	32-33	CCCD413C	Accumulated solar radiation:11.8 KJ
Register	34-35	999A4361	Solar radiation power: 225.6W/m2
Register	36-37	00004238	True wind direction: 46°
Register	38-39	F854440D	Altitude: 567.9 m

Register	40-41	63AC3CAE	True wind speed: 0.2 m/s
Register	42-43	00000000	Snow thickness(reserved)
Register	44-45	47AE3FE1	UV Radiation: 1.76 W/m2
Register	46-47	CCCD412C	PM1.0: 10.8 µg/m3
Register	48-49	00004198	PM10: 19.0 µg/m3
Register	50-51	00000000	Color Temp(reserved)
Ending Characters		8279	Checksum

Int type Take wind direction for example

D1	D0
Register 3 higher byte	Register 3 lower byte
00	38
higher byte	lower byte

Transformed as int type, value is **0x0038 => 56°**

- **Float type(IEEE754 Standard)** Take temperature for example

D3	D2	D1	D0
Register 7 higher byte	Register 7 lower byte	Register 6 higher byte	Register 6 lower byte
41	E7	33	33
higher byte	middle byte1	middle byte2	lower byte

Convert to float type, value is **0x41E73333 => 28.9** ℃

- **Appendix I. Device State 1 (Device State Sheet)**

BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT 10	BIT 9	BIT 8
1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0
Color Temp	UV Radiation	Accumulated Snow Thickness	True Wind Speed	True Wind Direction	Altitude	Visibility	Luminance
BIT 7	**BIT 6**	**BIT 5**	**BIT 4**	**BIT 3**	**BIT 2**	**BIT 1**	**BIT 0**
1/0	1/0	1/0	1/0	1/0	1/0	1/0	1/0
Solar Radiation	PM1.0 PM2.5 PM10	GPS	Precipitation	Compass Heading	Pressure	Relative Wind	Temperature Humidity

Note:Only when status bit is "1", corresponding data is valid, otherwise it's invalid. Same for below.

- **Appendix II. Precipitation State（HEX）**

BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT 10	BIT 9	BIT 8
0	0	0	0	0	0	0	0
BIT 7	**BIT 6**	**BIT 5**	**BIT 4**	**BIT 3**	**BIT 2**	**BIT 1**	**BIT 0**
0	0	0	0	0/1	solid 0/1	snow 0/1	rain 0/1

Clear accumulated precipitation Command(fixed string):

- **Request:**
  (HEX) 01 10 00 0F 00 02 04 00 00 00 00 B3 EF
- **Response:**
  (HEX) 01 10 00 0F 00 02 71 CB

Commands and Procedures

Following parameters such as communication address or baud rate, system time, precipitation automatic clear period can be set by users.

Commands		Content	Response
Instruction 1	ASCII	>*\r\n	>CONFIGURE MODE\r\n
	HEX	3E 2A 0D 0A	0A 3E 43 4F 4E 46 49 47 55 52 45 20 4D 4F 44 45 0D 0A
Remark	Enter Setting Mode		
Instruction 2	ASCII	>CUS 9600 8-N-1	>CMD IS SET
	HEX	3E 43 55 53 20 39 36 30 30 20 38 2D 4E 2D 31 0D 0A	3E 43 4D 44 20 49 53 20 53 45 54 0D 0A
Remark	Configure serial port configuration as Baud Rate 9600 bps; Data bits:8 bits; Parity:NONE; Stop bits:1 bit.		
Instruction 3	ASCII	>ID 2\r\n	>CMD IS SET
	HEX	3E 49 44 20 32 0D 0A	3E 43 4D 44 20 49 53 20 53 45 54 0D 0A
Remark	Configure address of device as 2. Inquiry address command is HEX: 3E 49 44 0D 0A		
Instruction 4	ASCII	>RESET\r\n	System start ok!\r\n
	HEX	3E 52 45 53 45 54 0D 0A	53 79 73 74 65 6D 20 73 74 61 72 74 20 6F 6B 21 0D 0A
Remark	Reboot device		
Instruction 5	ASCII	>!\r\n	>NORMAL MODE\r\n
	HEX	3E 21 0D 0A	3E 4E 4F 52 4D 41 4C 20 4D 4F 44 45 0D 0A
Remark	Exit setting mode to normal mode.		
Notice: 1. Characters “\r\n” is CRLF Carriage-Return Line-Feed, corresponding to HEX (0x0D,0x0A)			

Setting Procedures

No.	Function	Instructions
1	Set Communication Address	1→3→5→4
2	Set Serial Port Parameters	1→2→5→4

I .Appendix LRC Verification

A. Using a C language function code to generate LRC values

The function code uses 2 independent variables：

unsigned char *auchMsg; // To generate the LRC value, point the pointer to the buffer containing the binary data

unsigned short usDataLen; //Number of bytes in the buffer。

//This function returns LRC as a type“unsigned char” 。

```
// RC check code generation
static unsigned char LRCCheck(auchMsg, usDataLen)
unsigned char *auchMsg; /* calculating by information byte LRC*/
unsigned short usDataLen; /*calculating by information byteLRC*/
{
	unsigned char uchLRC = 0 ; /*Initializing LRC characters */
	while (usDataLen--) /*through the data buffer*/
	uchLRC += *auchMsg++; /*Add buffer byte Buffer byte no carry*/
	return ((unsigned char)(–((char)uchLRC))) ; /*reture to Binary complement*/
}
```

II .Appendix transform HEX to float data

Use C language's subfunction to transform 4 bytes(HEX) as float data(C language)

```
union
{
float    TestData_Float;
unsigned char TestArray[4];
}TData;
```

Analysis example:

D3	D2	D1	D0
Higher byte of register 2	Lower byte of register 2	Higher byte of register 1	Lower byte of register 1
40	AC	19	DF
Higher byte	Middle byte 1	Middle byte 2	Lower byte

After transformed to float data, value: 5.378

Subfunction:

```
float Tempfloat;
TData.TestArray [3]= 0x40; //input higher byte
TData.TestArray [2]= 0xac; //
TData.TestArray [1]= 0x19; //
TData.TestArray [0]= 0xdf; //input lower byte
Tempfloat = TData.TestData_Float;    //return result 5.378
```